Step 3: information and full-time sensitivity

14 October 2026 — analytical review under the approved H1/T1 family

Step 2 is user-approved and closed. This note derives review references without changing its equations, running traffic trajectories or selecting calibrated traffic parameters. It separates an exact model-relative information/action possibility from externally useful advice.

1. Fixed inputs and one analytical comparison family

Use the closed H1/T1 contract: E ends, A receives early transfers and B is the other continuing lane. Total road length is $L = L_p + b$, initial footprint $[0, \ell]$, and the terminal starts empty. The common action is $v \in [v_-, v_+]$; continuous mass is $M = \ell\rho$. Initial current continuing densities satisfy $a_0 = b_0$. The scalar early-preparation law in physical progress is

$$e_z = -\frac{1}{h}\min\{e, (\delta(v) + e)/\gamma\}, \quad \delta(v) = k_c(v) - r, \quad r = e_0 + a_0, \quad k_c(v) = \frac{wk_j}{v + w}. \tag{1}$$

The final domain has $v_+ r < R \le C(v_-)$, $v_+ b_0 \le R_B \le C(v_-)$ and $M < k_j b$. Consequently $r < k_c(v_+)$, $\delta(v) > 0$, and terminal ending sending does not bind the weighted server.

For reference speed $u > 0$ and density $\kappa > 0$, retain one shared geometry and parameter family:

$$\begin{gathered} v_- = u, \quad v_+ = 2u, \quad w = 2u, \quad k_j = 3\kappa, \quad \gamma = 2, \quad R = \frac{39}{20}\kappa u, \quad R_B = \kappa u, \\ b = \ell, \quad L_p = \ell + h\log 2, \quad L = 2\ell + h\log 2, \quad 0 < \eta = \ell/h < \log(4/3), \quad \alpha \ge 1. \end{gathered} \tag{2}$$

Compare two initial preparations:

	e_0	$a_0 = b_0$	r	ρ
$\mathcal{L}$	$3\kappa/5$	$3\kappa/10$	$9\kappa/10$	$6\kappa/5$
$\mathcal{H}$	$7\kappa/10$	$\kappa/4$	$19\kappa/20$	$6\kappa/5.$

(3)

All actions and both preparations satisfy the closed initial, ordinary-load, exit-capacity and storage bounds: $C(v_-) = 2\kappa u$ and $M/(k_j b) = 2/5$. Only initial ending mass differs as the one latent degree under symmetry. The late work parameter is shared across preparations and actions; no numerical α is selected for control experiments.

2. A reachable complete pre-decision aggregate history

The post-decision task still has no arrivals after time zero. To qualify initial reachability, introduce only this disclosed pre-decision loading history, using the same H1 local balance law and a common past speed u. Let $T = \ell/u$; the road is empty before $-T$. For each preparation, during $-T \le t \le 0$, prescribe upstream lane density traces

$$e_i^{\mathrm{in}}(t) = e_{0,i}\exp(-ut/h), \quad a_i^{\mathrm{in}}(t) = r_i - e_i^{\mathrm{in}}(t), \quad b_i^{\mathrm{in}}(t) = b_{0,i}. \tag{4}$$

They are exogenous preparation data, not another optimized control or future receiving trace. The common recipe and its parameters may be given to all predictors; the unobserved initial ending statistic selects the member.

Because $\exp(\eta) < 4/3$,

$$e_{\mathcal{H}}^{\mathrm{in}} < 14\kappa/15 < 19\kappa/20 = r_{\mathcal{H}}, \quad e_{\mathcal{L}}^{\mathrm{in}} < 4\kappa/5 < 9\kappa/10 = r_{\mathcal{L}}. \tag{5}$$

All input lane densities are nonnegative and below $k_c(u) = 2\kappa$. The early donor branch is feasible throughout loading since $a + 2e = r + e < 2\kappa$; the larger bound is $19/20 + 14/15 = 113/60 < 2$.

At time t, the occupied front is $x_f = u(t+T) \le \ell < L_p$. Within $0 < x < x_f$, the three lane densities equal the common time traces in (4); they solve the same-coordinate H1 transport balances. For example $e_t = -ue/h$, $e_x = 0$, exactly matching the donor-limited source. Boundary entries have those same traces, so each newly entering characteristic joins the common homogeneous body. At time zero this construction yields (3), the required uniform footprint and empty terminal. Initial continuing symmetry holds at the decision instant; the loading lane inputs are not separately asserted symmetric.

Give the aggregate observer the stronger lane-blind envelope of density, longitudinal flow, traffic speed and cumulative counts at every cross-section for the entire pre-decision history. In both preparations,

$$\begin{aligned} K_\Sigma(x,t) &= \rho\, \mathbf{1}_{0<x<u(t+T)}, \\ Q_\Sigma(x,t) &= uK_\Sigma(x,t), \\ N_\Sigma(x,t) &= \rho[\, u(t+T) - x\,]_+, \qquad -T \le t \le 0. \end{aligned} \tag{6}$$

Speed is u where traffic is present; empty-channel masks are identical. Earlier histories are empty in both cases, and all terminal counts are zero. Thus every standard lane-blind channel above, and any function or deterministic filter of it, is identical for all pre-decision times. A finite sensor/history subset cannot distinguish the pair if even this stronger envelope cannot.

Lane counts, lane inputs and lateral-event counts are additional channels, not secretly discarded longitudinal aggregate data. A lateral-event channel could identify the preparation under this recipe. If it is actually admitted, it must be given to A and its inference reassessed. This is a constructed analytical reachability certificate, not evidence that real or microsimulated data have these exact histories.

The shared pre-decision population-time prefix is $\int_{-T}^{0} \rho u(t+T)\, dt = MT/2$. It is identical and independent of the future action; excluding it from the decision-time objective changes no comparison. No action-dependent travel or waiting is removed.

3. The compact rival exactly closes prediction

Given M, ℓ and initial ending mass E_0,

$$e_0 = E_0/\ell, \qquad a_0 = b_0 = (M - E_0)/(2\ell), \qquad r = e_0 + a_0. \tag{7}$$

The closed law then predicts all early densities, terminal entry/eligibility histories, queue and departures. A+E and explicit lane representation G therefore have identical objectives and sensitivities under this member. Predicted histories do not constitute extra initial information. An aggregate-only deterministic predictor receives identical inputs in (6); an arbitrary learned prior can express uncertainty but cannot identify both actual preparations from these channels.

4. When preparation becomes action-dependent

For $\gamma = 2$ and $\delta > 0$, if $e_0 \le \delta$ the donor branch applies for all progress and $e = e_0 e^{-z/h}$. Otherwise the initial service branch switches to the donor branch at

$$z_s = 2h \log \frac{e_0 + \delta}{2\delta}, \qquad e(z; v) = \frac{(e_0 + \delta)^2}{4\delta} e^{-z/h} \quad (z \ge z_s). \tag{8}$$

Preparation $\mathcal{L}$ is donor-limited throughout every action because $\delta_{\mathcal{L}}(v) \geq 3\kappa/5 = e_{0,\mathcal{L}}$. Its completed preparation per initial position is action-independent.

For $\mathcal{H}$, the service branch starts when $v > v_c = 18u/11$. It ends before the first handoff exposure $z_0 = L_p - \ell = h \log 2$ throughout that interval. Indeed z_s is largest at v_+, where $(25/22)^2 < 2$. The front residual $B(v) = e(v, z_0)$ is

$$B_{\mathcal{H}}(v) = \begin{cases} 7\kappa/20, & v \leq v_c, \\ (e_{0,\mathcal{H}} + \delta_{\mathcal{H}})^2/(8\delta_{\mathcal{H}}), & v > v_c. \end{cases} \tag{9}$$

At fastest advice,

$$B_{\mathcal{H}}(v_+) = 125\kappa/352, \qquad B'_{\mathcal{H}}(v_+) = 225\kappa/(7744u) > 0, \qquad B_{\mathcal{L}} = 3\kappa/10, \quad B'_{\mathcal{L}} = 0. \tag{10}$$

The derivative in (10) is the feasible left derivative at the upper action bound. At v_c, preparation's first derivative is zero on both sides. Different residual merging alone is not yet a different speed optimum.

5. An exact full-time reference with sustained terminal waiting

Every handoff exposure in (2)–(3) is beyond the applicable initial service switch. Eligible lane-end arrivals begin at $t_f = (L - \ell)/v$ and last $d = \ell/v$. Writing $s = t - t_f$ during that pulse,

$$\dot{P}_E = vB(v)e^{-vs/h}, \quad \dot{P}_A = v[r - B(v)e^{-vs/h}], \quad \dot{P}_B = vb_0, \qquad 0 < s < d. \tag{11}$$

Define final forced mass and a pulse-area quantity

$$k = 1 - e^{-\eta}, \quad m(v) = hB(v)k, \quad J(v) = hB(v)\ell - hm(v), \quad D_0(v) = \ell r - Rd. \tag{12}$$

m is mass; J has units veh-eq m; D_0 has ordinary-work-equivalent mass units. On the branch with positive queue throughout the eligible pulse,

$$q_E(s) = \frac{(vr - R)s + (\alpha - 1)hB(v)(1 - e^{-vs/h})}{\alpha}, \qquad q_E(d) = \frac{D_0 + (\alpha - 1)m}{\alpha}. \tag{13}$$

After the pulse, only ending traffic remains and it drains at R/α. The exact queue area and complete TST are

$$\begin{aligned} Q_{\text{pulse}} &= \frac{D_0 d}{2\alpha} + \left(1 - \frac{1}{\alpha}\right)\frac{J}{v}, \\ Q_{\text{tail}} &= \frac{[D_0 + (\alpha - 1)m]^2}{2\alpha R}, \\ F(v) &= \frac{M(L - \ell/2)}{v} + Q_{\text{pulse}} + Q_{\text{tail}}. \end{aligned} \tag{14}$$

Travel of every class and the full clearance tail are counted. These are closed-form references, not numerically integrated traffic runs.

A simple sufficient branch condition for both preparations and all actions in this family is $\alpha > 17/3$. The donor bound gives ending density at the latest exposure at least $9\kappa/40$ and $r \geq 9\kappa/10$. Thus the whole pulse satisfies

$$v[r + (\alpha - 1)e] \geq u\kappa \left[\frac{9}{10} + (\alpha - 1)\frac{9}{40}\right] > R. \tag{15}$$

This branch restriction is not an empirical work-factor range.

6. A conditional information/action possibility, with an explicit limit

Rearrange (14) exactly as

$$\begin{aligned} F(v) &= \frac{\alpha m(v)^2}{2R} + I_0(v) + I_1(v)/\alpha, \\ I_0 &= \frac{M(L-\ell/2)+J}{v} + \frac{(D_0-m)m}{R}, \\ I_1 &= \frac{D_0 d}{2} - \frac{J}{v} + \frac{(D_0-m)^2}{2R}. \end{aligned} \tag{16}$$

For $\mathcal{H}$ at v_+, $A_+ = m(v_+)m'(v_+)/R > 0$. I_0', I_1' there are finite and independent of α. Consequently

$$F_{\mathcal{H}}'(v_+) = \alpha A_+ + I_0'(v_+) + I_1'(v_+)/\alpha > 0 \tag{17}$$

for the nonempty analytical parameter range

$$\alpha > \max\left\{\frac{17}{3}, \frac{|I_0'(v_+)| + |I_1'(v_+)|}{A_+}\right\}. \tag{18}$$

No numerical value is chosen from this range. Increasing the late work cost eventually amplifies the extra unfinished mass at faster advice enough to outweigh its travel advantage.

For $\mathcal{L}$, preparation is fixed across actions. The closed member's earlier-arrival dominance therefore makes faster advice no worse for its $A + E$ streams. Positive unqueued B mass has strictly decreasing travel time, so v_+ is its unique optimum. For $\mathcal{H}$, its preparation derivative is zero near v_-; the same strict travel improvement applies there. Under (18), fastest advice is not optimal because a feasible decrease reduces F. Continuity on the compact action interval therefore places its global minimizers strictly inside that interval; uniqueness is not asserted.

The pair has identical complete aggregate histories but disjoint optimal action sets, and opposite one-sided cost slopes at v_+. A common observable policy cannot be optimal for both. The initial ending scalar resolves this ambiguity and G adds no further predictive information.

Limit. This is a new conditional certificate within the approved family, not a calibrated work-factor result, practical gradient or empirical prevalence. The small-work result below is deliberately retained alongside it. Intermediate work factors and other admitted geometries need their own complete-objective classification; no positivity filter is authorized.

A common fastest decision for $1 \le \alpha \le 2$ in this reference family

This negative result does not assume sustained waiting or use (14) outside its branch. Split the local speed sensitivity into changing arrival timing with the prepared mass profile fixed, and changing that profile at fixed arrival times. The first part cannot increase $A + E$ time by earlier-arrival dominance. Positive unqueued B mass gives

$$F'_{\text{timing},\mathcal{H}} \le -\frac{b_{0,\mathcal{H}}\ell(L-\ell/2)}{v^2} < -\frac{\kappa h^2}{u^2}\frac{\eta}{32}, \tag{19}$$

using $v \le 2u$, $b_{0,\mathcal{H}} = \kappa/4$ and $\log 2 > 1/2$.

For a positive queue episode, onset is the first eligible arrival because weighted inflow decreases. At fixed clocks its preparation sensitivity is $(\alpha-1)\partial_{v,\text{prep}}P_E/\alpha$, bounded by $(\alpha-1)m'/\alpha$. The queue episode lasts at most $d + \alpha m/R$. A moving emptying endpoint adds no queue-area term because its queue is zero. With no queue the local preparation contribution is zero; the one-sided limit obeys the same bound.

In (9), $B'_{\mathcal{H}}$ increases from zero to its value in (10). Indeed, putting $K_0 = wk_j$, its positive-branch expression is $(e_0^2/\delta^2 - 1)(\delta + r)^2/(8K_0)$; its derivative with respect to δ is negative. Thus $m' \le hkB'(v_+)$. For $1 \le \alpha \le 2$, $k < \eta$, $\eta < 1/3$, $225/7744 < 1/32$ and $625/1716 < 1/2$ give

$$\begin{aligned} 0 \le F'_{\mathrm{prep},\mathcal{H}} &\le \frac{1}{2} m'_{\max} \left(\frac{\ell}{u} + \frac{2m_{\max}}{R} \right) \\ &< \frac{\kappa h^2}{u^2} \frac{3\eta^2}{128} < \frac{\kappa h^2}{u^2} \frac{\eta}{128}. \end{aligned} \tag{20}$$

Combining (19)–(20) makes complete $F_{\mathcal{H}}$ strictly decrease through the action interval, including the continuous event-boundary limits. $\mathcal{L}$ already has the same strict preference for every α. Both therefore uniquely prefer v_+ for $1 \le \alpha \le 2$. This classifies the history-matched subfamily, not every H1/T1 parameter set. Hidden preparation remains a prediction ambiguity but is not needed to choose the optimum there.

7. The sensitivity path and later numerical evidence

Each initial position x becomes eligible at $(L - x)/v$ before terminal delay. Let this time be $t_x(v)$; the cumulative arrival histories are

$$P_E(t; v) = \int_0^\ell e(v, L_p - x) \mathbf{1}_{\,t_x(v) \le t}\, dx, \quad P_A(t; v) = \int_0^\ell [r - e(v, L_p - x)] \mathbf{1}_{\,t_x(v) \le t}\, dx. \tag{21}$$

P_B uses constant density b_0. This retains both preparation sensitivity and moving arrival times. For interior pulse times, $x_{\min} = L - vt$,

$$\partial_v P_E = \int_{x_{\min}}^\ell e_v(v, L_p - x)\, dx + t\, e(v, vt - b). \tag{22}$$

Freezing travel lags or arrival endpoints would remove the second term. Holding initial traffic, geometry and work parameters fixed is part of the physical control derivative.

Let $W = P_A + \alpha P_E$. The ending-queue reflection is

$$q_E(t; v) = \frac{1}{\alpha} \left[W(t; v) - Rt - \min_{0 \le s \le t} \{W(s; v) - Rs\} \right]. \tag{23}$$

Eligible weighted inflow decreases during the pulse because e decreases. Therefore it admits at most one positive ending-queue excursion. The review event inventory comprises the possible preparation branch switch, first/last eligibility, queue onset/emptying and the pulse-end/service change. Grazing or coincident events are not assumed smooth. Capacity and spillback boundaries are admission boundaries, not extra clipped branches.

Choose a common fixed integration horizon larger than $L/v_- + \alpha M/R$ for each fixed parameter case. All mass has cleared there. Inside a differentiable event regime,

$$\frac{dF}{dv} = -\frac{M(L - \ell/2)}{v^2} + \int_0^{H_*} \partial_v q_E(t; v)\, dt. \tag{24}$$

Moving queue endpoints contribute no separate queue-area boundary term because $q_E = 0$ there, but their timing effects must already enter queue sensitivity through (21)–(23). The final population is zero, so the clearance endpoint does not license dropping the tail.

Later execution must first verify forward conservation/admission/clearance and A+E/G equality. Compare the continuous analytical reference, the numerical forward model's directional differences and its AD derivative separately. Internal AD/difference agreement is not continuous-model

convergence. Use double precision, a stable feasible step ladder, recorded event signatures and numerical forward/derivative convergence. At action bounds use feasible one-sided tests. Report events separately; neither smoothing nor removal of forward boundary cases is authorized.

Numerical tolerances, time/solver resolution and perturbation sizes must be fixed in a later approved execution plan using the reference conditioning and error budget. No arbitrary values or urban defaults are transplanted here.

8. External evidence and present disposition

The proof of matched histories belongs to H1's conditional loading construction. It is not automatically a matched-history proof for another traffic model. The large-work range (18) has no independent empirical admission. An independently coded use of (1), (14) or (23) is a correctness reference, not traffic-validity or independent-usefulness evidence.

SUMO's official speed-sign, vehicle-behavior, insertion, state and trip-output documentation was screened. It supports a potential operational speed-limit reference with distinct vehicle interactions; it does not supply this task's population realization, exact observer match, terminal handoff/priority, work-factor calibration or complete evaluation mapping. A native continuous lane-aware source model is another candidate, with different response and service qualifications. Neither reference is adopted here.

The Step 3 source/evaluator audit and conceptual report carry the genuine remaining choice: which independent reference meaning to qualify first. Step 2 remains closed. Step 3 has substantive analytical findings and a draft evidence protocol, but its full evaluation contract is not frozen. No simulator, dependency installation, traffic experiment, AD or optimization run was performed.